JACK A. NOMINA

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EDUCATION

Miami University, Oxford, OH | *MS* Mechanical Engineering | *BA* Music Performance | *BS* Mechanical Engineering
Graduated Undergrad May 2026 | GPA: 3.92 | GRE: 169 Qualitative, 167 Quantitative, 5 Writing

ACCOLADES AND ACHIEVEMENTS

- **Ohio Space Grant Consortium Best Poster (2025)** – Selected as best poster from over 50 undergraduate researchers from across Ohio. Poster received a perfect score from evaluation by a research professional.
- **President's List** – 2 semesters | **Dean's List** – 4 semesters
- **TSA 2022 Engineering Design National Champion** - Competed against hundreds of total teams nationally, prompted to design a solution to "Restore or improve urban infrastructure". Worked on a team of 6 to design and build a semi-automated aquaponic farm. Project involved extensive technical writing, and efficiency improved from 30-90% over 3 testing phases.
- **Eagle Scout** - Lead fellow scouts through various leadership roles. Organized and completed an Eagle Scout Project.

RESEARCH EXPERIENCE

SMART MATERIALS LAB, Dr. Jeong-Hoi Koo, Miami University — June 2024 – Present

- Leveraged flexible silicone-based materials alongside the smart material known as Electro-Rheological Fluid to create a flexible haptic module. Tested various prototype modules using a Dynamic Mechanical Analyzer (DMA).
- Design, fabrication, and testing of a Tuned Vibration Absorber that leverages the capabilities of Electro-Rheological Fluid. This project will be completed during the 2025-26 academic year.

ADVANCED ROBOTICS AND MECHATRONICS LAB, Konkuk University — June 2024 – August 2024
International Mechanical Engineering Research Intern

- Designed, fabricated, and tested a haptic module using the smart material Electro-Rheological Fluid
- Improved testing efficiency by 500% over the course of one week, drastically improving the turn-around time on the evaluation of prototype haptic module effectiveness.
- Presented research results weekly to an audience of non-native English speakers, effectively communicating data and important discoveries despite a language barrier.

PUBLICATIONS AND PRESENTATIONS

UNDERGRADUATE SUMMER SCHOLARS PRESENTATION — August 2024

- Presenting research on haptic systems to an audience with extensive research experience but zero knowledge of smart materials or haptics. Short presentation window (verbal accompanied by slideshow) followed by questions from peers and professors

OHIO SPACE GRANT CONSORTIUM PRESENTATION — April 2025

- Presenting research on haptic systems to an audience of other student researchers, professors, and NASA scientists at NASA Glenn Research Center. Audience had STEM research experience but did not necessarily have smart materials experience. Presentation via a poster and accompanying verbal explanation

OHIO SPACE GRANT CONSORTIUM PRESENTATION — April 2026

- Presenting research on the design of a tuned vibration absorber to an audience of other student researchers, professors, and NASA scientists at NASA Glenn Research Center. Audience had STEM research experience but did not necessarily have smart materials experience. Presentation format was slideshow accompanied by explanation

ELECTRO-RHEOLOGICAL FLUID TUNED VIBRATION ABSORBER — In Progress

- Preparing a journal paper on the design and testing of a tuned vibration absorber that employs the smart material electro-rheological fluid in order to alter the damping coefficient and spring constant of the absorber.
- The goal of this study is to create an adaptable tuned vibration absorber that dampens different resonant frequencies depending on the resonant frequency of objects that it is placed on.
- Intended to submit for review in Summer 2026

SKILLS

CAD Software (Autodesk Inventor, Fusion360, AutoCAD, SolidWorks), Machine Learning, Neural Networks, Machine Design, Advanced Mechanics of Materials, Measurement Systems, Control Systems, Instrument Calibration, MATLAB Sensor Integration, FEMM Analysis Software, Project Management, Ladder Logic, Python, MATLAB, C#, C++, Unity Engine, Technical Writing, Engineering Design and Drawings, Conversational Spanish

WORK EXPERIENCE

MIAMI UNIVERSITY, Oxford, OH — February 2026 – May 2026

Grader (MME311 Dynamics)

- Modified answer keys to ensure correctness and readability
- Graded homework assignments for accuracy on a tight timeline while maintaining fairness amongst students

AGARVEY, LLC, Xenia, OH — May 2025 – August 2025

Engineering Intern

- Modified and upgraded a mask-cleaning system to clean CBRNE equipment for military and first responder use, tested/presented product at military conferences such as CBOA. Improved clean quality while reducing system propensity to overheat
- Created a bill of materials, CAD models, wrote code for product. Wrote funding applications such as Phase II SBIRs.

THE FISCHER GROUP, Fairfield, OH — June 2022 - August 2022

Mechanical Engineering Intern

- Collaborated with fellow interns to design and fabricate manufacturing clamps and oil distribution systems.
- Worked with pneumatic systems and PLC ladder logic code to create an automated system allowing line workers to replace materials without halting line production, saving up to 2 minutes with each replacement.
- Gained proficiency during a short period in manufacturing and fabrication methods such as CAM software and subtractive manufacturing with metals.

LEADERSHIP

ASME – MIAMI UNIVERSITY STUDENT CHAPTER

President (2025)

- Planning bi-weekly meetings to promote professional development amongst mechanical engineering majors
- Coordinating with faculty to connect students to professional and research opportunities

Vice President (2024)

- Assisted president in presidential duties
- Completed university funding requirements for the organization

Secretary (2023)

- Maintained club organization
- Communicated club event details to general body members

ENGINEERS WITHOUT BORDERS – MIAMI UNIVERSITY STUDENT CHAPTER

Event Planning Chair (2025)

- Organizing and facilitating all aspects of two large-scale events for the student chapter
- Planning the 2025 Great Lakes Regional Convention, attended by many local student and professional chapters
- Planning the 2025 Fall Fundraising Gala with a target profit of $5000 and projected attendance of over 120 students and professionals.

Vice President (2024)

- Facilitated communication between multiple student and professional chapters
- Ensured all project and operation teams met goals along expected timelines
- Represented the chapter at various events, streamlined administrative processes

Local Projects Manager (2023)

- Led a team of ~15 student volunteers in progressing the completion of a steel truss bridge
- Organized weekly meetings to keep general body members engaged in the project
- Taught non-engineering majors how to perform truss calculations

THETA TAU – TAU DELTA CHAPTER
Departmental Liaison (Spring 2025)
- Represented the organization in university advertising content weekly
- Planned the 2025 Miami University Senior Design Expo for the College of Engineering and Computing

Vice President (2024)
- Managed 20+ event chairs across the chapter, ensuring weekly events were held and goals were met
- Ensured that the organization received adequate university funding
- Trained members in leadership styles

Professional Development Co-Chair (Fall 2023)
- Led weekly events with the goal of improving the professional knowledge of members
- Planned and facilitated a mixer event for all members to meet company representatives

MIAMI UNIVERSITY MME STUDENT LEADERSHIP COUNCIL
Member (2025-2026 Academic Year)
- Planned events to increase engagement of mechanical and manufacturing engineering students with the department
- Coordinated with students and professors to improve departmental relations and communications